

BONAIRE INTL, Bonaire, Sint Eustatius and SabaWMO: **789900**Lat: **12.131N**Lon: **68.269W**Elev: **23**StdP: **14.68**Time Zone: **-4.00 (W04)**Period: **09-19**WBAN: **00413****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	73.8	75.1	67.4	100.6	79.8	68.1	103.4	79.6	26.8	83.6	25.4	83.3	7.4	80	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	8.8	90.7	80.0	89.9	79.7	89.5	79.5	81.9	87.4	81.0	87.2	80.5	87.0	14.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	158.4	85.8	79.1	151.4	85.3	78.8	149.8	85.2	45.6	88.4	44.8	87.9	44.2	87.0	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	23.9	22.7	DB	71.1	93.2	1.3	1.4	70.1	94.2	69.3	95.0	68.6	95.8	67.6	96.8	(4)
(5)			WB	68.7	84.4	1.3	3.7	67.8	87.1	67.0	89.2	66.2	91.3	65.2	93.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	80.9	80.8	81.7	82.8	83.9	84.4	84.3	85.2	85.5	84.9	83.4	81.9
	DBStd	2.14	1.39	1.34	1.34	1.24	1.11	1.03	1.12	1.20	1.61	1.72	2.00	1.62
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12162	957	862	982	984	1050	1032	1064	1093	1064	1081	1003	990
	CDD65	6687	492	442	517	534	585	582	599	628	614	616	553	525
	CDH74	75653	4635	4069	5116	5704	6722	6931	7167	7859	7874	7639	6409	5528
CDH80	26440	963	852	1233	1637	2309	2630	2726	3411	3609	3248	2334	1488	
Wind	WSAvg	14.9	14.5	15.4	15.6	15.8	16.9	17.5	16.4	14.9	13.7	12.6	11.0	14.1
Precipitation	PrecAvg	20.7	1.9	0.8	0.6	0.7	0.6	0.7	1.2	0.9	1.6	2.9	5.6	3.3
	PrecMax	39.6	5.8	3.5	3.0	3.7	3.2	3.5	4.4	2.9	8.0	9.4	16.4	12.2
	PrecMin	10.6	0.1	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.1	0.1	1.4	0.2
	PrecStd	7.8	1.3	0.8	0.7	0.9	0.9	0.8	1.0	0.7	1.7	2.4	3.5	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	87.5	87.7	88.2	89.7	90.0	90.0	91.4	91.8	91.2	90.0	88.2
		MCWB	76.9	77.8	77.1	78.6	79.9	80.2	79.7	80.1	80.5	79.6	80.1	78.4
	2%	DB	85.9	85.8	86.4	87.7	88.2	88.7	88.7	90.0	90.9	90.0	89.2	87.1
		MCWB	76.5	75.6	76.5	78.1	79.0	79.1	79.1	79.5	80.0	79.6	79.4	78.0
	5%	DB	84.7	84.6	85.9	86.7	87.8	88.0	88.1	89.5	89.9	89.5	88.1	86.2
		MCWB	75.9	75.3	76.3	77.4	78.8	78.9	78.9	79.4	79.8	79.5	79.1	77.7
	10%	DB	84.3	84.3	84.9	86.1	86.9	87.5	87.6	88.6	89.4	88.4	87.5	85.8
		MCWB	75.8	75.1	75.8	76.9	78.3	78.8	78.8	79.3	79.7	79.5	78.9	77.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.2	79.0	79.0	80.7	81.0	82.2	83.0	83.7	82.8	82.3	81.4	80.1
		MCDB	84.2	85.4	85.1	86.3	87.4	87.0	87.7	89.4	87.9	87.6	87.4	86.3
	2%	WB	78.1	77.5	78.0	79.7	80.0	80.1	80.1	80.9	81.7	81.1	80.3	79.4
		MCDB	83.7	83.7	84.2	85.3	86.3	86.7	86.5	87.5	87.5	87.1	86.9	85.4
	5%	WB	77.3	76.5	77.3	78.9	79.6	79.8	79.8	80.1	81.0	80.5	79.9	78.6
		MCDB	83.3	82.7	83.7	84.6	85.7	86.3	86.1	86.8	87.3	86.8	86.2	84.4
	10%	WB	76.5	76.0	76.7	78.2	79.1	79.2	79.2	79.8	80.4	80.0	79.4	78.0
		MCDB	82.6	82.2	83.2	84.0	85.0	85.7	85.5	86.4	86.9	86.2	85.4	83.8
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.4	8.3	7.8	7.4	7.4	7.8	8.3	8.8	8.3	8.7	8.2
		MCDBR	8.2	8.6	8.5	8.0	7.8	7.7	8.0	8.6	8.9	8.6	8.9	8.5
		MCWBR	4.1	3.6	3.5	3.2	3.4	3.7	3.9	4.0	3.9	3.7	3.9	4.3
	5% WB	MCDBR	8.1	7.8	8.0	7.6	7.8	7.6	7.9	8.4	8.8	8.3	8.7	8.1
		MCWBR	4.7	3.9	4.1	3.7	4.0	4.4	4.5	4.3	4.5	4.0	4.2	4.5
Clear-Sky Solar Irradiance	taub	0.387	0.409	0.420	0.486	0.476	0.505	0.484	0.441	0.426	0.402	0.392	0.395	
	taud	2.554	2.468	2.446	2.238	2.301	2.199	2.279	2.440	2.474	2.574	2.589	2.559	
	Ebn at Noon	291	288	285	264	262	252	258	272	279	286	287	285	
	Edh at Noon	32	36	37	46	42	46	43	37	36	32	31	31	
All-Sky Solar Radiation	RadAvg	1681	1866	1972	1979	1960	1996	2026	2063	1987	1770	1578	1569	
	RadStd	68	58	99	75	85	76	56	83	76	95	159	78	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	-0.59	-0.43	-0.86	-0.86	N/A	N/A	N/A	N/A

Nomenclature: See separate page